

RAG Phone Project — Retrieval-Augmented Generation for Mobile Device Q&A

One-Page Executive Summary | Academic Course Submission

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1. PROJECT OVERVIEW

This project implements a complete end-to-end **Retrieval-Augmented Generation (RAG) pipeline** designed to answer natural-language questions about mobile phone manuals and specifications. The system combines **hybrid retrieval** — BM25 sparse indexing fused with dense semantic embeddings — with a **cross-encoder reranker** to maximise both recall and precision before generation. Inference is supported via both the **OpenAI cloud API** (gpt-3.5-turbo / gpt-4) and a **local Ollama (LLaMA 2)** endpoint, surfaced through a real-time Streamlit web UI.

2. System Components

Component	Description
app.py	Streamlit UI; routes queries, displays streamed LLM responses
ingestion/pipeline.py	Document loading, chunking (512 tokens, 64 overlap), embedding & BM25 indexing
hybrid_retriever.py	BM25 + cosine similarity; Reciprocal Rank Fusion; cross-encoder reranking
requirements.txt	Python deps: requests, numpy, streamlit, sentence-transformers, rank-bm25, openai

4. How to Run `cd C:\Users\malha\Documents\rag_phone_project python -m venv venv && venv\Scripts\activate pip install -r requirements.txt ollama run llama2 (local LLM mode) streamlit run app.py` Open `http://localhost:8501`

3. Key Technical Highlights **Hybrid Retrieval:** BM25 sparse + dense semantic embeddings fused via Reciprocal Rank Fusion for high-recall candidate selection. **Cross-Encoder Reranking:** Retrieved passages rescored for precision before being passed to the LLM generator. **Dual LLM Support:** OpenAI API (gpt-3.5-turbo / gpt-4) and local Ollama REST API with token-by-token streaming. **Live Streaming UI:** Streamlit streams JSON response frames in real-time; confirmed final answer: *"Hello! My name is LLaMA..."*

5. Known Issues & Fixes

Issue	Fix
Missing requests / numpy	<code>pip install requests numpy</code>
Streamlit stale cache	Clear from menu or call <code>st.cache_data.clear()</code>
Wrong Python interpreter	Confirm venv activated; check which python

6. NEXT STEPS

- Add **RAGAS evaluation metrics** (faithfulness, context recall) for systematic pipeline benchmarking.
- Implement **semantic / sentence-level chunking** to improve context boundary quality over fixed-token chunking.
- **Dockerize** the full stack and deploy to cloud infrastructure (Azure Container Apps or HuggingFace Spaces).